

EDCI 59100-016: Educational Research and Literature Review

# Week 3 Comprehensive Lecture Notes

Finding, Reading, and Evaluating Literature

Summer 2026

**Major themes covered:** database searching, Boolean search strings, review-method alignment, search documentation, source screening, source evaluation, skimming scholarly articles, reference management software, AI-supported research workflows, discussion preparation, and the Search Results and Source Evaluation Matrix.

Prepared from the saved Week 3 course materials in the local Brightspace course-content folder.

# Contents

|                                                             |           |
|-------------------------------------------------------------|-----------|
| <b>1 Learning Objectives</b>                                | <b>3</b>  |
| <b>2 Topic Overview</b>                                     | <b>3</b>  |
| 2.1 How Week 3 Connects to Previous Work . . . . .          | 3         |
| 2.2 Real-World Relevance . . . . .                          | 4         |
| <b>3 The Week 3 Research Workflow</b>                       | <b>4</b>  |
| <b>4 Searching Within Databases</b>                         | <b>4</b>  |
| 4.1 Purpose of the Search Stage . . . . .                   | 4         |
| 4.2 Choosing Databases . . . . .                            | 5         |
| 4.3 Database Searching Is Testing, Not Finalizing . . . . . | 5         |
| <b>5 Building Search Strings</b>                            | <b>5</b>  |
| 5.1 Why Search Strings Matter . . . . .                     | 5         |
| 5.2 Step-by-Step Search String Construction . . . . .       | 5         |
| 5.3 Example Search Strings . . . . .                        | 6         |
| 5.4 How Databases Read the Search . . . . .                 | 6         |
| <b>6 Inclusion and Exclusion Criteria</b>                   | <b>6</b>  |
| 6.1 Purpose . . . . .                                       | 6         |
| 6.2 Keep, Maybe, Exclude . . . . .                          | 7         |
| <b>7 Method-Specific Search Expectations</b>                | <b>7</b>  |
| 7.1 Narrative Review . . . . .                              | 7         |
| 7.2 Integrative Review . . . . .                            | 7         |
| 7.3 Theoretical Review . . . . .                            | 8         |
| 7.4 Scoping Review . . . . .                                | 8         |
| 7.5 Systematic Review . . . . .                             | 8         |
| <b>8 Citation Chaining and Snowballing</b>                  | <b>9</b>  |
| <b>9 Practicing Search Refinement</b>                       | <b>9</b>  |
| 9.1 How to Judge Improvement . . . . .                      | 10        |
| 9.2 Method-Specific Refinement Questions . . . . .          | 10        |
| <b>10 Documenting the Search</b>                            | <b>10</b> |
| 10.1 The Search Log . . . . .                               | 10        |
| 10.2 Search Log Quality . . . . .                           | 11        |
| <b>11 Screening Candidate Sources</b>                       | <b>11</b> |
| 11.1 Initial Screening Notes . . . . .                      | 11        |
| 11.2 How Method Changes Screening . . . . .                 | 12        |
| <b>12 Evaluating Sources</b>                                | <b>12</b> |
| 12.1 When Evaluation Happens . . . . .                      | 12        |
| 12.2 Big 5 Criteria . . . . .                               | 12        |
| 12.3 CRAAP Test . . . . .                                   | 13        |
| 12.4 Initial Evaluation Questions . . . . .                 | 13        |

|                                                                         |           |
|-------------------------------------------------------------------------|-----------|
| 12.5 Deeper Evaluation During Reading . . . . .                         | 13        |
| <b>13 Reading Scholarly Articles Efficiently</b>                        | <b>13</b> |
| 13.1 Why Strategic Reading Matters . . . . .                            | 13        |
| 13.2 Recommended Reading Sequence . . . . .                             | 14        |
| 13.3 The Outside-In Article Strategy . . . . .                          | 14        |
| <b>14 Skimming and Scanning</b>                                         | <b>14</b> |
| 14.1 Skimming . . . . .                                                 | 14        |
| 14.2 Scanning . . . . .                                                 | 15        |
| 14.3 A Practical Skimming Procedure . . . . .                           | 15        |
| <b>15 Reference Management Software</b>                                 | <b>15</b> |
| 15.1 Why Use a Citation Manager . . . . .                               | 15        |
| 15.2 Recommended Organization . . . . .                                 | 15        |
| 15.3 Connection to Later Writing . . . . .                              | 16        |
| <b>16 AI Tools and Algorithmic Thinking</b>                             | <b>16</b> |
| 16.1 What AI Can Support . . . . .                                      | 16        |
| 16.2 What AI Cannot Replace . . . . .                                   | 16        |
| <b>17 Assignment Focus: Search Results and Source Evaluation Matrix</b> | <b>16</b> |
| 17.1 What the Assignment Should Demonstrate . . . . .                   | 16        |
| 17.2 Required Components . . . . .                                      | 17        |
| 17.3 Evaluation Criteria . . . . .                                      | 17        |
| <b>18 Worked Example: Refining a Search</b>                             | <b>17</b> |
| 18.1 Step 1: Identify Concept Groups . . . . .                          | 18        |
| 18.2 Step 2: Build a Broad Search . . . . .                             | 18        |
| 18.3 Step 3: Add Context . . . . .                                      | 18        |
| 18.4 Step 4: Screen Sample Results . . . . .                            | 18        |
| <b>19 Source Evaluation Matrix</b>                                      | <b>18</b> |
| <b>20 Common Misconceptions</b>                                         | <b>19</b> |
| <b>21 Important Notes for Studying</b>                                  | <b>19</b> |
| <b>22 Summary</b>                                                       | <b>19</b> |
| <b>23 Review Questions</b>                                              | <b>20</b> |
| 23.1 Conceptual Questions . . . . .                                     | 20        |
| 23.2 Short-Answer Practice With Answers . . . . .                       | 20        |
| 23.3 Application Questions . . . . .                                    | 20        |
| <b>24 Cheat Sheet</b>                                                   | <b>21</b> |
| <b>25 Source Materials Used</b>                                         | <b>21</b> |

# 1 Learning Objectives

After completing Week 3, students should be able to:

1. Implement a search strategy using discipline-appropriate databases and structured search strings based on the Week 2 literature review plan.
2. Translate research questions into searchable concept groups, keywords, synonyms, Boolean operators, quotation marks, and database-specific syntax.
3. Document the search process transparently in a search log, with the amount of documentation matched to the selected review methodology.
4. Screen candidate articles and make initial Keep, Maybe, or Exclude decisions using explicit relevance, quality, and method-fit criteria.
5. Evaluate sources using criteria such as currency, relevance, authority, accuracy, purpose, methodology, evidence quality, bias, and contribution to the literature.
6. Use skimming and scanning strategies to decide which scholarly articles deserve deeper reading.
7. Organize sources in reference management software so that later synthesis and writing are more efficient.
8. Explain how AI tools can support, but not replace, transparent searching, reading, screening, and documentation.

## 2 Topic Overview

Week 3 marks a shift from designing a literature review plan to executing the early stages of the review. In Week 2, the emphasis was on clarifying a topic, selecting a review type, and planning a search. In Week 3, the emphasis moves to action: running searches, testing search terms, comparing databases, documenting decisions, screening candidate sources, and beginning to evaluate which studies are genuinely useful.

This topic matters because a literature review is only as credible as the process that produced it. A strong paper does not simply contain many sources. It shows that the researcher searched deliberately, selected sources for defensible reasons, evaluated evidence critically, and aligned every decision with the review methodology.

**Key takeaway.** Searching is not a one-time task. It is an iterative research process. You test terms, examine results, revise the search, document what changed, and keep track of why articles were kept, marked as possible, or excluded.

### 2.1 How Week 3 Connects to Previous Work

The Week 2 literature review plan provides the starting point:

- The research question defines the main concepts to search.
- The review type defines how broad, selective, or systematic the search should be.
- The initial search plan suggests databases, keywords, and possible criteria.
- The purpose statement clarifies what kinds of evidence will matter.

Week 3 then tests whether the plan actually works in databases. If a search returns thousands of unrelated records, the plan needs refinement. If a search returns almost nothing, the plan may be too narrow or may use terminology that the literature does not use. If the first relevant studies use different keywords than expected, those keywords should be added to the search log and tested.

## 2.2 Real-World Relevance

Transparent literature searching is essential in graduate research, academic publishing, grant writing, curriculum design, health research, engineering reviews, policy work, and industry research. A project team may need to know what has already been tried, what evidence is strong, where results conflict, and where gaps remain. These decisions depend on the same skills practiced this week: searching, documenting, screening, evaluating, and organizing sources.

**Figure placeholder: Literature review workflow**

A process diagram should show the flow from research question to concept groups, databases, search strings, search log, screening, source evaluation matrix, preliminary included studies, and later synthesis. The diagram should also show feedback loops from screening back to revised keywords and search strings.

## 3 The Week 3 Research Workflow

The recommended weekly flow is:

1. Search within databases for the topic.
2. Read and evaluate candidate literature.
3. Explore reference management software and citation managers.
4. Apply algorithmic thinking when using AI tools for literature review work.
5. Complete the discussion on screening decisions and search challenges.
6. Complete the Search Results and Source Evaluation Matrix.
7. Submit the final literature review plan.

These tasks are connected. The search log records what was done. The screening table records what was found and what decision was made. The source evaluation matrix records what useful information each source contributes. The discussion gives a place to explain search challenges, compare strategies, and reflect on how method affects decisions.

## 4 Searching Within Databases

### 4.1 Purpose of the Search Stage

The purpose of Week 3 searching is to apply the plan to actual databases. At this stage, students are not expected to have the final perfect corpus of sources. They are expected to test and improve the process.

Before beginning, gather:

- the research question or purpose statement;
- the list of keywords and synonyms from the literature review plan;
- the selected review type;
- the initial database plan;
- any inclusion and exclusion criteria already drafted.

**Important.** The search process should be treated as evidence. A search log is not busywork. It helps show how the literature was located, how the strategy changed, and whether the review is methodologically credible.

## 4.2 Choosing Databases

Different topics belong in different scholarly ecosystems. A database is not just a search box; it is a collection with its own disciplinary coverage, indexing rules, controlled vocabulary, filters, and search syntax.

| Topic area                     | Useful databases                                  | Why they fit                                                                                                                                              |
|--------------------------------|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Education                      | ERIC, PsycINFO                                    | ERIC focuses on education research; PsycINFO captures psychology, learning, cognition, motivation, behavior, and educational psychology.                  |
| Technology and computing       | IEEE Xplore, ACM Digital Library                  | These databases capture computer science, engineering, systems, artificial intelligence, human-computer interaction, and technical conference literature. |
| Health and medicine            | PubMed                                            | PubMed is central for biomedical and health sciences literature and supports structured medical searching.                                                |
| Broad interdisciplinary topics | Scopus, Web of Science, Google Scholar, OneSearch | These tools capture research across fields and can be useful when a topic crosses disciplinary boundaries.                                                |

To decide where to search, ask:

- Where is research in this field typically published?
- Does the database include the journals, conferences, or disciplines that matter for the topic?
- Do the articles already found cite sources from certain journals or fields repeatedly?
- Do Purdue library subject guides recommend discipline-specific databases?
- If uncertain, what happens when one discipline-specific database is compared with Google Scholar?

## 4.3 Database Searching Is Testing, Not Finalizing

The first database choice is not always the final choice. A database may be too broad, too narrow, too technical, too clinical, or too weak for a topic. Students should test databases by examining the first page or two of results and asking whether the records consistently align with the research question.

For example, a search about AI feedback in college writing may produce useful education studies in ERIC, psychology-oriented studies in PsycINFO, technical writing studies in Scopus, and computer-science model papers in ACM or IEEE. The correct choice depends on whether the review is about teaching practice, learner outcomes, writing assessment, model design, or the broad state of the field.

# 5 Building Search Strings

## 5.1 Why Search Strings Matter

Academic databases do not understand a full research question the way a human reader does. They match words, phrases, metadata, subject terms, abstracts, titles, authors, and indexed fields. A search string translates the research question into a form the database can process.

If a search is too simple, it can return too many unrelated results. If a search is too narrow, it can miss important studies. Search strings help control the search by combining the major concepts of the research question.

## 5.2 Step-by-Step Search String Construction

1. Identify two or three key concepts from the research question.

2. List synonyms, abbreviations, and related terms for each concept.
3. Combine similar terms with OR.
4. Combine different concept groups with AND.
5. Use quotation marks for exact phrases.
6. Use parentheses to group terms.
7. Test the string in a database and inspect the first 5 to 10 results.
8. Revise one element at a time so you can tell what improved or weakened the search.

### 5.3 Example Search Strings

For a topic about AI feedback in writing, possible strings include:

```
("artificial intelligence" OR AI) AND (feedback OR "automated feedback") AND writing
```

```
AI AND writing AND ("learning outcomes" OR performance)
```

The first string is broader because it includes multiple feedback terms. The second is narrower because it adds outcomes or performance. Neither string is automatically correct. Each must be tested against the review purpose.

### 5.4 How Databases Read the Search

Most academic databases treat search syntax as formal instructions:

- AND requires both concept groups to appear.
- OR accepts any term within a group of similar terms.
- Quotation marks keep words together as a phrase.
- Parentheses control how terms are grouped.
- Field filters such as title, abstract, subject, or full text can change results dramatically.
- Small syntax changes can create large changes in the number and quality of records.

**Common mistake.** Do not type the entire research question into a database and expect precise scholarly results. Break the question into searchable concepts and controlled combinations.

#### Figure placeholder: Boolean search logic

A diagram should show three concept circles: AI terms, writing terms, and feedback terms. The overlap where all three circles intersect represents the effect of AND. Smaller circles inside each concept group represent synonyms joined by OR.

## 6 Inclusion and Exclusion Criteria

### 6.1 Purpose

Inclusion and exclusion criteria define what counts as a relevant source before the researcher becomes too attached to individual articles. Criteria help make screening consistent and defensible.

Initial criteria may include:

- source type, such as empirical study, theoretical article, review article, dissertation, conference paper, or grey literature;
- population, such as K-12 students, higher education students, teachers, patients, workers, or organizations;
- context, such as online learning, construction sites, healthcare settings, engineering classrooms, or workplace training;
- time frame, such as the last 5 years or 2015 to 2025;
- method, such as qualitative, quantitative, mixed-methods, design-based research, experimental, or conceptual;
- language, publication status, peer-review status, and geographic scope;
- connection to the chosen review type.

## 6.2 Keep, Maybe, Exclude

Early screening often uses three practical categories:

|                |                                                                                                                      |
|----------------|----------------------------------------------------------------------------------------------------------------------|
| <b>Keep</b>    | The source clearly matches the topic, criteria, and review purpose.                                                  |
| <b>Maybe</b>   | The source is related but has an uncertain fit. It might differ by population, method, field, time period, or focus. |
| <b>Exclude</b> | The source does not fit the topic, criteria, review type, or quality expectations.                                   |

A Maybe source is not a failure. It is a normal part of early searching. In fact, Maybe sources are useful because they reveal boundary problems. If many Maybe sources are about a different population, the search may need more specific population terms. If many Maybe sources are technical model papers but the review is about learning outcomes, the search may need education-specific terms.

## 7 Method-Specific Search Expectations

The selected review methodology changes how the search should be planned, documented, and judged. The same article can be a strong Keep for one review type and a Maybe or Exclude for another.

### 7.1 Narrative Review

A narrative review explains a topic by drawing together important, relevant, and often influential literature. It does not aim to retrieve every possible study.

**Search emphasis:**

- Begin broad, then refine toward studies that explain the topic clearly.
- Include foundational works and recent research.
- Use citation chaining to identify influential works.
- Prioritize sources that help tell the scholarly story of the topic.

**Documentation expectation:** Narrative reviews do not require rigid exhaustive search protocols, but the writer should still explain how key studies were identified.

**Guiding question:** What studies help explain this topic clearly?

### 7.2 Integrative Review

An integrative review brings together multiple forms of evidence and perspectives. It may include quantitative studies, qualitative studies, theoretical or conceptual work, and different methodological traditions.

**Search emphasis:**



- Use multiple search strings and databases.
- Intentionally capture variation in study types and perspectives.
- Avoid collecting many similar studies while missing conceptual or methodological diversity.
- Look for patterns across heterogeneous sources.

**Documentation expectation:** Document enough of the search process to show how variation across study types and perspectives was intentionally captured.

**Guiding question:** What range of evidence is needed to understand patterns across studies?

### 7.3 Theoretical Review

A theoretical review focuses on ideas, concepts, models, frameworks, and the development of theoretical relationships. It may include fewer sources than other review types, but each source should be influential and conceptually important.

**Search emphasis:**

- Search for key concepts, theorists, models, frameworks, and foundational terms.
- Prioritize seminal works and high-impact theoretical papers.
- Do not limit the search only to empirical studies.
- Use phrase searching, Boolean operators, proximity searching, and document-type filters when available.

**Documentation expectation:** Explain how selected works represent the development of the theory or concept.

**Guiding question:** What works help explain how this idea, theory, or framework developed?

### 7.4 Scoping Review

A scoping review maps the range, nature, and extent of literature on a topic. It is especially useful when a field is emerging, interdisciplinary, conceptually varied, or not yet ready for a narrow systematic review.

**Search emphasis:**

- Search broadly across databases, journals, conferences, and source types.
- Capture different populations, contexts, methods, and study types.
- Include grey literature when it is relevant to the review purpose.
- Use citation chaining to expand from seed articles.
- Consider search updates because literature can become outdated quickly.

**Documentation expectation:** PRISMA-ScR expects exact search strings, dates of searches, number of results per database, and a flow of records through identification, screening, and inclusion.

**Guiding question:** What has been studied, in what ways, and what has not been studied?

### 7.5 Systematic Review

A systematic review follows a predefined, transparent, and replicable process. It usually has narrow criteria and aims to answer a focused question.

**Search emphasis:**

- Define search terms in advance.
- Search multiple databases.
- Use clear inclusion and exclusion criteria.
- Record how many records were found, removed, screened, excluded, and included.

- Consider grey literature to reduce publication bias.
- Translate search strings for each database rather than pasting the same syntax everywhere.

**Documentation expectation:** Full documentation is required: database names, interfaces, search dates, full search strings, modifications, limits, filters, searcher information, results, screening counts, full-text exclusions, and final included studies.

**Guiding question:** Can someone else follow this process step by step and reach a comparable result set?

| Review type | Search goal                                         | Best evidence to prioritize                                              | Documentation level                          |
|-------------|-----------------------------------------------------|--------------------------------------------------------------------------|----------------------------------------------|
| Narrative   | Explain the topic through key literature            | Foundational works, recent studies, influential perspectives             | Moderate; explain strategy and key choices   |
| Integrative | Compare patterns across varied evidence             | Quantitative, qualitative, theoretical, and conceptual sources           | Moderate to high; show intentional variation |
| Theoretical | Trace concepts, frameworks, and models              | Seminal theoretical works and conceptual papers                          | Moderate; explain conceptual lineage         |
| Scoping     | Map the extent and range of literature              | Broad sources across contexts, populations, methods, and grey literature | High; PRISMA-ScR style tracking              |
| Systematic  | Answer a focused question with replicable searching | Studies meeting predefined criteria                                      | Very high; PRISMA-style tracking             |

## 8 Citation Chaining and Snowballing

Citation chaining helps identify literature that may not appear through keyword searches alone.

### Backward snowballing

Looking at the references of a relevant seed article to find earlier foundational studies.

### Forward snowballing

Looking up who has cited a seed article in Google Scholar, Scopus, or Web of Science to find newer work that built on it.

Snowballing is useful in all review types, but its role varies. A narrative or theoretical review may use it to find influential or foundational sources. A scoping or systematic review may use it as a supplementary search method that must be documented.

**Exam tip.** Be able to explain the difference between backward and forward snowballing and why both can improve a literature search.

## 9 Practicing Search Refinement

The Week 3 practice task asks students to compare and refine a search.

1. Choose one search string.
2. Run two searches: one in a university database and one in Google Scholar.
3. Compare the first 5 to 10 results.

4. Ask what types of sources appear.
5. Ask whether the sources are similar or different.
6. Decide which search is more useful for the topic.
7. Make one change: adjust a keyword, add specificity, or refine the focus.
8. Run the search again and judge whether it improved.

## 9.1 How to Judge Improvement

Improvement does not always mean fewer results. It means better alignment between the search output and the review purpose.

Ask:

- Do the results consistently relate to the topic?
- Do they reflect the selected review method?
- Do they begin to show patterns across studies?
- Are the same irrelevant topics appearing repeatedly?
- Are key expected concepts missing?
- Does the search capture both foundational and recent work when needed?

## 9.2 Method-Specific Refinement Questions

- Narrative review: Are these studies key and influential? Do they help explain the topic clearly?
- Integrative review: Do the results include different types of studies, or are they too similar?
- Theoretical review: Are the results finding concepts, frameworks, and foundational ideas, or mostly empirical reports?
- Scoping review: Do the results reflect a range of contexts, populations, and study types?
- Systematic review: Can the inclusion or exclusion of each result be justified by predefined criteria?

# 10 Documenting the Search

## 10.1 The Search Log

A search log records the search process in a transparent way. It helps the researcher remember what was tried, avoid repeating ineffective searches, explain decisions, and demonstrate methodological rigor.

A strong search log includes:

- search number;
- database or platform used;
- exact search string;
- filters or limits applied;
- number of results;
- notes on relevance, problems, and decisions.

---

| # | Database | Search string | Filters | Results | Notes |
|---|----------|---------------|---------|---------|-------|
|---|----------|---------------|---------|---------|-------|

---

|   |                |                                                                                   |                                                                 |                                                     |
|---|----------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------|
| 1 | ERIC           | ("artificial intelligence" OR AI) AND feedback AND writing                        | Peer-reviewed; last 1,245 10 years                              | Too broad; included K-12 and higher education.      |
| 2 | ERIC           | ("artificial intelligence" OR AI) AND feedback AND writing AND "higher education" | Peer-reviewed; last 312 10 years                                | More focused; better population alignment.          |
| 3 | Google Scholar | Same core concepts adapted to platform                                            | Relevance sorting Record or date filters as count needed varies | Useful for citation tracing, but less reproducible. |

## 10.2 Search Log Quality

A weak search log only lists what was searched. A strong search log explains what the search taught the researcher. For example, the note “too broad” is useful, but it becomes stronger when it identifies why the search was too broad: wrong population, wrong field, too many technical sources, too many opinion pieces, too many unrelated uses of a term, or missing method language.

**Important.** Always record exact strings before editing them. Once a useful search is changed, it can be difficult to reconstruct what produced the earlier result set.

## 11 Screening Candidate Sources

### 11.1 Initial Screening Notes

Screening asks whether each source belongs in the developing review. At the early stage, screening does not require a full detailed reading of every article. It uses titles, abstracts, keywords, headings, introductions, methods, conclusions, and sometimes reference lists to decide whether the source deserves further attention.

A screening table should include:

- article title;
- Keep, Maybe, or Exclude decision;
- reason for the decision;
- explanation of method fit.

| Article title                                    | Decision | Why                                                                | Fits method?                                                                                  |
|--------------------------------------------------|----------|--------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| AI Feedback in College Writing Courses (2022)    | Keep     | Direct match to topic; higher education focus; empirical findings. | Integrative review: provides empirical evidence that can be compared with other study types.  |
| AI Tools for K-12 Writing Support (2019)         | Maybe    | Relevant topic but different population.                           | Integrative review: may add variation across contexts if the review includes K-12 comparison. |
| General AI Productivity Tools in Business (2021) | Exclude  | AI topic but not education, writing, feedback, or target context.  | Does not fit the review purpose.                                                              |

## 11.2 How Method Changes Screening

The same source can be evaluated differently depending on the review type.

| Method      | Likely decision | Reasoning                                                                                                                                 |
|-------------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Narrative   | Keep            | The article explains key ideas clearly and is widely cited or highly relevant.                                                            |
| Integrative | Keep            | The article adds empirical evidence that can be compared with qualitative, theoretical, or conceptual work.                               |
| Theoretical | Maybe           | The article is useful only if it connects to a theory, model, framework, or conceptual debate.                                            |
| Scoping     | Keep            | The article represents one context, population, method, or dimension within the broader literature landscape.                             |
| Systematic  | Keep or Exclude | The decision depends strictly on predefined criteria such as population, design, date range, intervention, outcome, and publication type. |

## 12 Evaluating Sources

### 12.1 When Evaluation Happens

Source evaluation happens at several stages:

1. when the source first appears in search results;
2. when the abstract and introduction are skimmed;
3. when the source is read more carefully;
4. when it is compared with other sources during synthesis;
5. when it is finally used in the literature review.

The goal is to ensure the review uses literature that is relevant, accurate, credible, and appropriately interpreted.

### 12.2 Big 5 Criteria

The Big 5 criteria are:

**Currency** When was the source published? Is it current enough for the field?

**Coverage/Relevance** How closely does it connect to the research question?

**Authority** Who wrote it? Are the authors credible for this topic?

**Accuracy** Are claims supported by evidence? Are there obvious errors?

**Objectivity/Purpose** Does the source have a bias, agenda, or purpose that affects interpretation?

For medical, scientific, and technology topics, sources from the last 5 years are often preferred because knowledge changes quickly. For less time-sensitive humanities or historical topics, sources from the last 5 to 10 years may be acceptable, and older foundational sources may remain essential.

## 12.3 CRAAP Test

The CRAAP Test is a mnemonic that overlaps with the Big 5:

|                  |                                                                                |
|------------------|--------------------------------------------------------------------------------|
| <b>Currency</b>  | Is the source current enough?                                                  |
| <b>Relevance</b> | Does it answer or inform the research question?                                |
| <b>Authority</b> | Is the author or organization credible?                                        |
| <b>Accuracy</b>  | Are the claims evidence-based and verifiable?                                  |
| <b>Purpose</b>   | Why was the source created? To inform, persuade, sell, entertain, or advocate? |

## 12.4 Initial Evaluation Questions

During the first pass, ask:

- When was the source published?
- Is it relevant to the topic?
- What are the author qualifications?
- Is the source scholarly or peer-reviewed?
- Are sources cited to support claims?
- Does the journal, website, or organization have a clear bias?
- Does the summary or introduction use emotional or biased language?
- Are there obvious spelling, grammar, or factual problems?

## 12.5 Deeper Evaluation During Reading

Once a source passes initial evaluation, read it with more precise questions:

1. Is there visible bias in the work?
2. How was the research conducted?
3. What are the strengths and weaknesses of the methodology?
4. How does the author justify the conclusions?
5. What similarities does this source share with other articles?
6. How does this source differ from other articles?
7. What new knowledge, evidence, method, or insight does it contribute?
8. What unanswered questions or future research opportunities does it leave?

**Common mistake.** A peer-reviewed article is not automatically strong for every literature review. It still needs to fit the question, method, criteria, and purpose of the review.

# 13 Reading Scholarly Articles Efficiently

## 13.1 Why Strategic Reading Matters

Literature review work requires reading many sources under time constraints. Reading every article from the first word to the last word is often inefficient during early screening. Strategic reading helps decide which sources deserve deep attention.

## 13.2 Recommended Reading Sequence

The saved reading strategy materials recommend skipping around at first:

1. Read the title and abstract.
2. Skim the introduction.
3. Jump to the conclusion or discussion.
4. Look at results, figures, tables, charts, graphs, or images.
5. If the source is relevant, then read it from start to finish.
6. Take notes, summarize sections, and record citation information.
7. Check the reference list for additional related sources.

This sequence works because the abstract, introduction, discussion, and conclusion usually explain the article's purpose, context, findings, and contribution. The methods and results are critical for careful evaluation, but they may not be the best starting point during early relevance screening.

## 13.3 The Outside-In Article Strategy

Scholarly articles often put the big-picture information on the outside:

- abstract;
- introduction;
- discussion;
- conclusion.

The technical core is often inside:

- methodology;
- instruments;
- data collection;
- statistical analysis;
- results details.

The outside sections help decide relevance. The inside sections help evaluate validity, replicability, and strength of evidence.

## 14 Skimming and Scanning

### 14.1 Skimming

Skimming means reading quickly to understand the main ideas, structure, and argument without focusing on every word. It helps researchers decide whether a source is relevant enough for deeper reading.

Types of skimming include:

|                           |                                                                                                    |
|---------------------------|----------------------------------------------------------------------------------------------------|
| <b>Preview skimming</b>   | Looking at titles, headings, abstracts, and introductions before deciding whether to read closely. |
| <b>Overview skimming</b>  | Moving through the full article quickly to understand its structure, flow, and argument.           |
| <b>Selective skimming</b> | Looking for specific terms, sections, or concepts related to the research question.                |

**Comprehensive skimming**

Combining speed with light analysis by noticing evidence, examples, and data while avoiding full deep reading.

**14.2 Scanning**

Scanning means looking for specific information such as dates, names, measures, populations, theories, keywords, or statistics. Skimming gives the big picture. Scanning locates precise details.

**14.3 A Practical Skimming Procedure**

1. Start with the title and abstract. Ask whether the source aligns with the research question.
2. Read the introduction and conclusion to understand the purpose and outcome.
3. Review headings and subheadings to understand organization.
4. Examine visuals and data displays.
5. Read the first and last sentences of important paragraphs.
6. Mark keywords, definitions, theories, and method terms.
7. Review references for possible citation chaining.

**Key takeaway.** Skimming is not lazy reading. It is a screening strategy. Deep reading should be reserved for sources that pass initial relevance and quality checks.

**15 Reference Management Software****15.1 Why Use a Citation Manager**

Reference management software supports the literature review workflow by storing citations, PDFs, notes, tags, folders, metadata, and citation exports. It reduces the risk of losing sources and helps organize evidence before writing.

Common tools include:

- Zotero;
- Mendeley;
- EndNote;
- RefWorks;
- database export files such as RIS and BibTeX.

**15.2 Recommended Organization**

For Week 3, a practical source library can include:

- a folder for all raw search exports;
- a folder for Keep sources;
- a folder for Maybe sources;
- a folder for Excluded-after-screening sources if the review requires tracking;
- tags for method, population, context, theory, database, and screening decision;
- notes that summarize why each source matters.



## 15.3 Connection to Later Writing

The goal of reference management is not just citation formatting. The deeper purpose is retrieval. When writing the literature review, the researcher needs to quickly find studies by theme, method, finding, population, or limitation. A well-organized library makes synthesis easier.

# 16 AI Tools and Algorithmic Thinking

## 16.1 What AI Can Support

AI tools can help with parts of the literature review workflow:

- brainstorming synonyms and concept groups;
- translating a research question into possible search strings;
- summarizing long abstracts or articles;
- identifying recurring concepts across notes;
- checking whether a screening rationale is clear;
- generating draft tables for organization;
- creating plain-language explanations of difficult methods.

## 16.2 What AI Cannot Replace

AI tools cannot replace the researcher's responsibility to:

- verify every source against the original article;
- record the actual database search string and result count;
- decide inclusion and exclusion based on the selected review method;
- evaluate methodology and evidence quality;
- avoid fabricated citations or unsupported claims;
- preserve transparency and replicability.

**Important.** AI output should be treated as a draft aid, not as evidence. The evidence is the database record, article, method, results, and documented screening decision.

# 17 Assignment Focus: Search Results and Source Evaluation Matrix

The Week 3 assignment asks students to submit a working Searching and Evaluating Literature document. The assignment focuses on documenting and reflecting on the process for identifying and evaluating literature.

## 17.1 What the Assignment Should Demonstrate

The assignment should show that the student is:

- developing an effective search strategy;
- applying inclusion and exclusion criteria;
- making thoughtful screening decisions;
- beginning to identify patterns across sources;
- revising the process based on feedback and search results.

## 17.2 Required Components

1. **Search Log:** databases used, search strings, filters, number of results, and notes on search effectiveness.
2. **Screening Decisions:** sources reviewed, Keep/Maybe/Exclude decisions, and brief explanations tied to criteria.
3. **Preliminary Set of Included Studies:** sources likely to remain in the review, even if the final set changes.
4. **Source Evaluation Matrix:** article notes that include main idea, method, key findings, and why the source matters.

The expected scope is at least 10 sources across screening and evaluation. These sources may include Keep, Maybe, and Exclude decisions. The purpose is to show process and reasoning, not merely to collect a high number of citations.

## 17.3 Evaluation Criteria

The assignment is evaluated on:

- quality of the search process;
- clarity of screening decisions and reasoning;
- development of source evaluation notes;
- use of criteria and method alignment;
- overall clarity and organization.

| Criterion                     | Excellent                                                         | Proficient                                | Developing                              | Points |
|-------------------------------|-------------------------------------------------------------------|-------------------------------------------|-----------------------------------------|--------|
| Search log                    | Multiple searches documented with refinement and thoughtful notes | Searches documented with some explanation | Limited searches or minimal explanation | 5      |
| Screening decisions           | Clear Keep/Maybe/Exclude decisions using criteria                 | Decisions provided with some reasoning    | Decisions unclear or inconsistent       | 5      |
| Source set and evaluation     | Robust set of about 10 or more sources with developed notes       | Adequate set and some detail              | Fewer sources or limited evaluation     | 5      |
| Criteria and method alignment | Decisions clearly connect to criteria and review method           | Some connection present                   | Connection limited or unclear           | 3      |
| Clarity and organization      | Well organized and easy to follow                                 | Generally clear                           | Somewhat difficult to follow            | 2      |

## 18 Worked Example: Refining a Search

Suppose the research question is:

How does AI-supported automated feedback affect writing development in higher education?

## 18.1 Step 1: Identify Concept Groups

| Concept  | Possible terms                                                          |
|----------|-------------------------------------------------------------------------|
| AI       | artificial intelligence, AI, machine learning, automated, generative AI |
| Feedback | feedback, automated feedback, formative feedback, writing feedback      |
| Writing  | writing, composition, academic writing, writing instruction             |
| Context  | higher education, college, university, undergraduate, graduate          |

## 18.2 Step 2: Build a Broad Search

("artificial intelligence" OR AI) AND (feedback OR "automated feedback") AND writing

This string may be too broad because it can retrieve K-12 studies, general tool discussions, or non-education writing contexts.

## 18.3 Step 3: Add Context

("artificial intelligence" OR AI) AND (feedback OR "automated feedback") AND writing AND ("higher education" OR college OR university)

This version is more focused. If it removes too many important studies, the researcher can test whether some articles use “postsecondary” instead of “higher education” or whether a database subject heading captures college students more effectively.

## 18.4 Step 4: Screen Sample Results

The first 10 results should be inspected for source type, population, topic match, method, and review-method fit. The researcher should not silently choose only the articles that look easy to use. Every Keep, Maybe, or Exclude decision should be explainable.

# 19 Source Evaluation Matrix

A source evaluation matrix transforms articles into usable synthesis notes.

| Article   | Main idea                                   | Method        | Key findings                                          | Why it matters                               |
|-----------|---------------------------------------------|---------------|-------------------------------------------------------|----------------------------------------------|
| Article A | AI feedback system in college writing       | Mixed methods | Students valued speed, but quality varied by task     | Useful for feedback benefits and limitations |
| Article B | Automated writing evaluation in K-12        | Quantitative  | Scores improved, but population differs from target   | Maybe source for context comparison          |
| Article C | Conceptual framework for AI writing support | Theoretical   | Defines feedback timing, agency, and revision support | Useful for theoretical framing               |

Good matrix notes are brief but specific. They should not copy the abstract. They should extract the information needed for the literature review: what the study did, what it found, how strong the evidence is, and how it connects to the review purpose.

## 20 Common Misconceptions

**More sources means better review.**

Quality, fit, method alignment, and synthesis matter more than raw count.

**Google Scholar is enough.**

Google Scholar is useful, especially for citation chasing, but it is less controlled and less reproducible than discipline databases.

**A perfect search exists.**

Searching is iterative. A documented, reasonable, well-refined process is the goal.

**Maybe sources are useless.**

Maybe sources clarify boundaries and can reveal whether criteria need refinement.

**AI summaries can replace reading.**

AI summaries must be verified against the original article and cannot replace evaluation of methods and evidence.

**All review types require the same search.**

Narrative, integrative, theoretical, scoping, and systematic reviews have different search and documentation expectations.

## 21 Important Notes for Studying

**Important.** Review methodology controls search behavior. Always ask what the review type requires before judging whether the search is strong.

**Exam tip.** Know the difference between AND and OR. AND narrows by requiring multiple concepts. OR broadens by allowing synonyms or related terms within the same concept group.

**Common mistake.** Do not apply filters without documenting them. Date limits, language limits, peer-reviewed filters, and subject filters can all change the evidence base.

**Key takeaway.** The Week 3 deliverable is a working document. Sources and decisions can change later, but the current process should be clear, organized, and defensible.

## 22 Summary

Week 3 develops the practical mechanics of literature review work. Students move from a plan to a documented search process. They choose databases based on field and topic, build search strings from concept groups, test Boolean logic, compare database outputs, refine keywords, define inclusion and exclusion criteria, and begin screening sources.

The selected review method determines the search goal. Narrative reviews prioritize key explanatory works. Integrative reviews intentionally include varied evidence. Theoretical reviews trace concepts and frameworks. Scoping reviews map the range of literature. Systematic reviews require a predefined, transparent, and replicable process.

Source evaluation happens at multiple points. The Big 5 and CRAAP criteria help judge currency, relevance, authority, accuracy, and purpose. Deeper reading adds questions about bias, methodology, conclusions, similarities, differences, contributions, and research gaps.

Strategic reading is essential. Skimming, scanning, and outside-in article reading help researchers decide which sources deserve close attention. Reference management software supports organization, and AI tools can assist with brainstorming, summarizing, and structuring, but the researcher remains responsible for verification and documentation.

## 23 Review Questions

### 23.1 Conceptual Questions

1. Why is a search log important for a literature review?
2. How does a search string differ from a research question?
3. Why does OR usually broaden a search while AND usually narrows it?
4. What is the difference between initial screening and deeper source evaluation?
5. How does a scoping review search differ from a narrative review search?
6. Why might the same article be Keep for one review type and Maybe for another?
7. What are the Big 5 criteria for source evaluation?
8. How does the CRAAP Test help evaluate sources?
9. Why should AI-generated summaries be checked against the original article?
10. What is the difference between backward and forward snowballing?

### 23.2 Short-Answer Practice With Answers

**Question 1.** A student searches AI writing feedback and gets thousands of unrelated results. What are two improvements?

**Answer.** The student can group synonyms with Boolean operators, such as:

```
("artificial intelligence" OR AI) AND (feedback OR "automated feedback") AND  
writing
```

The student can also add context terms such as "higher education" or filter by peer-reviewed sources and date range if those limits match the review criteria.

**Question 2.** Why should a systematic review document database-specific syntax changes?

**Answer.** Different databases use different field codes, subject headings, truncation rules, and proximity operators. Documenting syntax changes makes the process transparent and helps another researcher understand how the same conceptual search was adapted across platforms.

**Question 3.** A source is relevant to the topic but uses a different population. Should it automatically be excluded?

**Answer.** No. It may be marked Maybe. The decision depends on the review method and criteria. An integrative or scoping review may use the source for variation or mapping, while a systematic review may exclude it if the population does not meet predefined criteria.

### 23.3 Application Questions

1. Build a search string for a research question about virtual reality training and teacher preparation.
2. Create three possible inclusion criteria and three exclusion criteria for a review about AI tutoring in higher education.

3. Choose one article from your own search results and justify a Keep, Maybe, or Exclude decision.
4. Compare how a narrative review and scoping review would treat the same broad topic.
5. Create a one-row source evaluation matrix for one source you found this week.

## 24 Cheat Sheet

| Concept              | Rule of thumb                                                           |
|----------------------|-------------------------------------------------------------------------|
| Database choice      | Match the database to the discipline and topic.                         |
| Search string        | Convert research questions into concept groups and synonyms.            |
| AND                  | Narrows by requiring different concepts together.                       |
| OR                   | Broadens by including synonyms or related terms.                        |
| Quotation marks      | Keep phrases together, such as "automated feedback".                    |
| Parentheses          | Group terms so the database reads the logic correctly.                  |
| Inclusion criteria   | Define what belongs in the review.                                      |
| Exclusion criteria   | Define what does not belong in the review.                              |
| Keep                 | Clearly fits topic, criteria, and method.                               |
| Maybe                | Related but uncertain fit; useful for boundary decisions.               |
| Exclude              | Does not fit topic, criteria, method, or quality expectations.          |
| Big 5                | Currency, coverage/relevance, authority, accuracy, objectivity/purpose. |
| CRAAP                | Currency, relevance, authority, accuracy, purpose.                      |
| Backward snowballing | Search the references of seed articles.                                 |
| Forward snowballing  | Search newer articles that cited seed articles.                         |
| Scoping review       | Map the range of evidence and gaps.                                     |
| Systematic review    | Follow a predefined, transparent, replicable process.                   |
| Skimming             | Read quickly for structure and main ideas.                              |
| Scanning             | Search for specific information.                                        |
| Reference manager    | Store, tag, organize, annotate, and cite sources.                       |

## 25 Source Materials Used

These notes were prepared from the saved Week 3 course-content materials, including the local Week 3 overview, "Searching Within Databases for Your Topic," "Reading and Evaluating Literature," "Tool Intro - Reference Management Software," "AI Tools - AI Algorithmic Thinking for Literature Review," the discussion and assignment references visible in the Week 3 module, the local assignment prompt for the Search Results and Source Evaluation Matrix, and the saved external reading pages on evaluating sources and reading/skimming scholarly articles.